



Drone Data Analysis Activity

Activity:

- Run and validate a drone script, monitor performance with GCS software
- Assess models through testing, data collection, and result documentation.

Materials Required:

- **Drone Simulation Software:** Software like Gazebo, AirSim, or PX4 SITL
- **Ground Control Station (GCS) Software:** QGroundControl or Mission Planner
- **Python:** For scripting and data analysis

Instructions:

1. Set up the Simulation Environment:

- Install the chosen drone simulation software on your computer.
- Create a simulated environment (e.g., indoor space, outdoor field) where the drone will operate.

2. Write a Drone Script:

- Use Python to write a script that defines the drone's flight path, actions, and parameters (e.g., altitude, speed, heading).
- Incorporate different mission types (e.g., waypoints, grid search, object tracking).

3. Load the Script into GCS:

- Open your GCS software and load the script you created.
- Configure the GCS settings to match the simulation environment.



4. Run the Simulated Flight:

- Start the simulation and execute the script from the GCS.
- Observe the drone's behaviour in the simulated environment.

5. Monitor Performance:

- Use the GCS to monitor the drone's real-time performance, including:
 - Altitude, speed, heading
 - Battery level
 - GPS signal strength
 - Sensor data (e.g., IMU, camera)

6. Collect and Analyse Data:

- Record the simulation data (e.g., flight logs, sensor readings, images).
- Use Python to analyse the collected data and assess the drone's performance.
 - Evaluate the accuracy of the flight path.
 - Assess the effectiveness of the script and mission types.
 - Analyse sensor data for anomalies or issues.

7. Document Results:

- Prepare a report summarising your findings, including:
 - Script details
 - Simulation setup
 - Performance metrics
 - Data analysis results
 - Challenges and recommendations